Predictive Statistical Problem

Alzheimer

Group #5

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Predictive Analytics

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Statement of Purpose

The main objective of this project is to determine whether different physical attributes, diseases, medical conditions, habits, among many others contribute to the likelihood of a person suffering from Alzheimer disease. This is very relevant because as it is a widely known condition, Alzheimer is a disorder that affects the correct functionality of the patient's memory and thinking skills. According to MayoClinic (2025) "About 6.9 million people in the United States age 65 and older live with Alzheimer's disease. Among them, more than 70% are age 75 and older." These huge numbers imply that a lot of people are afflicted by Alzheimer and their life quality gets worse. Moreover, this illness has no cure which means that the people suffering from it have problems in their everyday activities and knowing which physical conditions, diseases or characteristics make a person more likely to have this condition is a good way in doing something to prevent it.

Scope of the Project

This project wants to incorporate a logistic model to determine the likelihood of a patient having Alzheimer's disease based on various characteristics. The dataset contains multiple variables belonging to different areas related to the analysis. The categories and its variables are as follow:

a) Demographic Details (Age, Gender, Ethnicity and Education Level).
b) Lifestyle Factors (BMI, Smoking, Alcohol Consumption, Physical Activity, Diet Quality, and Sleep Quality.)
c) Medical History (Family History of Alzheimer, Cardiovascular Disease, Diabetes, Depression, Head Injury, and Hypertension.)
d) Clinical Measurements (Systolic and Diastolic Blood Pressure, Total Cholesterol, LDL and HDL Cholesterol, and Triglycerides Cholesterol.)

e) Cognitive and Functional Assessments (MMSE, Functional Assessment, Memory Complaints, Behavioral Problems, and ADL.)

f) Symptoms of Alzheimer (Confusion, Disorientation, Personality Changes, Difficulty Completing Tasks, and Forgetfulness.)

There are also columns belonging to unique categories such as Patient ID, Diagnosis of Alzheimer and a Doctor ID variable that won't be used for this analysis as it is the same for all rows.

The deliverables for this project include doing data preprocessing, exploratory analysis of the variables, model development, model evaluation and its interpretation with conclusions. The analysis will be focused on identifying significant predictors and assessing their impact on Alzheimer's diagnosis. The project is limited to only using the available dataset for analysis and conclusions. Finally, the output should be helpful in aiding with early detection and risk assessment for Alzheimer's patients, but it is not intended to be used for clinical purposes.

Background Research

Alzheimer's disease (AD) is the most common type of dementia, affecting at least 27 million people and corresponding from 60 to 70% of all dementia cases typically manifests through a progressive loss of episodic memory and cognitive function, subsequently causing language and visuospatial skills deficiencies, which are often accompanied by behavioral disorders such as apathy, aggressiveness and depression. Approximately 70% of the risk of developing AD can be attributed to genetics. However, acquired factors such as cerebrovascular diseases, diabetes, hypertension, obesity and dyslipidemia increase the risk of AD development. As a complement, some protective factors associated with a lower risk of disease incidence, such as cognitive reserve, physical activity and diet will also be addressed. (Silva et al., 2019)

According to Centers for Disease Control and Prevention (CDC) data, AD is ranked as the seventh leading cause of death in the United States in 2022. Significant progress has been made in developing biomarkers for specific and early diagnosis of AD over the past decade. These biomarkers include neuroimaging markers obtained through amyloid and tau PET scans, cerebrospinal fluid (CSF), and plasma markers, such as amyloid, tau, and phospho-tau levels.

There is no cure for AD, although there are treatments available that may alleviate and manage some of its symptoms. In recent years, there have been significant advancements in the development of medications that aim to moderate the progression of the disease, particularly with the discovery of new disease biomarkers. (Kumar et al., 2024)

Design and Data Collection Methods

The dataset was extracted from Kaggle from the following link:

https://www.kaggle.com/datasets/rabieelkharoua/alzheimers-disease-dataset

As this dataset is pre-collected it is considered a secondary data collection method.

For the analytical approach, a logistic model is the most appropriate choice of model as the dependent variable is binary and it needs to calculate the probability of a person having Alzheimer given different conditions in their physical traits, medical history, symptoms of possible diagnosis, etc. Moreover, it needs to consider the logistic model assumptions which will be considered in this analysis.

Methodology/Strategies

Data preprocessing is the first step to describe the characteristics of the variables as well as looking for outliers if there are. Also, it is important to mention that this dataset contains no missing values. Next, the exploratory analysis will consist of looking for relationships between the variables using correlation and choose the most important predictors for the dependent variable which will be Diagnosis of Alzheimer. Then, a logistic regression model will be done to classify patients as having or not having Alzheimer's. This model performance will be evaluated using metrics such as accuracy, precision, recall, F1 score and the area under the ROC curve. Testing statistical significance can be done to assess the individual impact of the variables in the outcome. Finally, conclusions and recommendations will be given using the results from the model.

Business Impact and Conclusion

The most notable benefit that this project wants to do is to look for any patterns or trends when knowing what different variables intervene when determining the likelihood of a person suffering Alzheimer. We hope that this project helps people know what factors they can look out for if someone in their families may have Alzheimer. Furthermore, it is important to mention that further research must be done to solidify the outcome of this project.

As in any study, it is necessary to know the composition of the data, look for outliers and possible values outside the range. Therefore, the following interpretations are attached for some important variables in the study.

Demographic Details

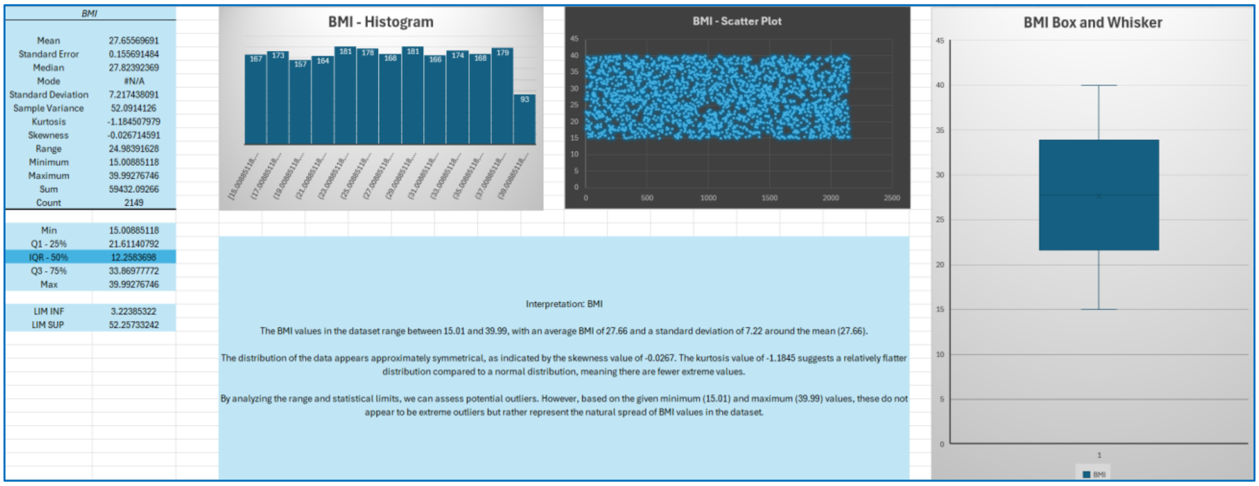
Age: The age of the patients ranges from 60 to 90 years.

Interpretation: The age of individuals in the dataset ranges between **60** and **90** years, with an average age of 74.91 years and a standard deviation of 8.99 years around the mean (74.91). From different statistical measures, we can infer that the data distribution is approximately symmetrical, though not perfectly, as indicated by the skewness value of 0.0459. Additionally, the kurtosis value of -1.1892 suggests a relatively flatter distribution compared to a normal distribution. By calculating the upper and lower limits and analyzing potential outliers, we can determine whether extreme values exist. However, based on the range and given values, the minimum (60) and maximum (90) ages may not necessarily be outliers but rather represent the natural span of the dataset.

Datasource : Sheet Age - excel file: Interpretations.xls

BMI: Body Mass Index of the patients, ranging from **15** to **40.**

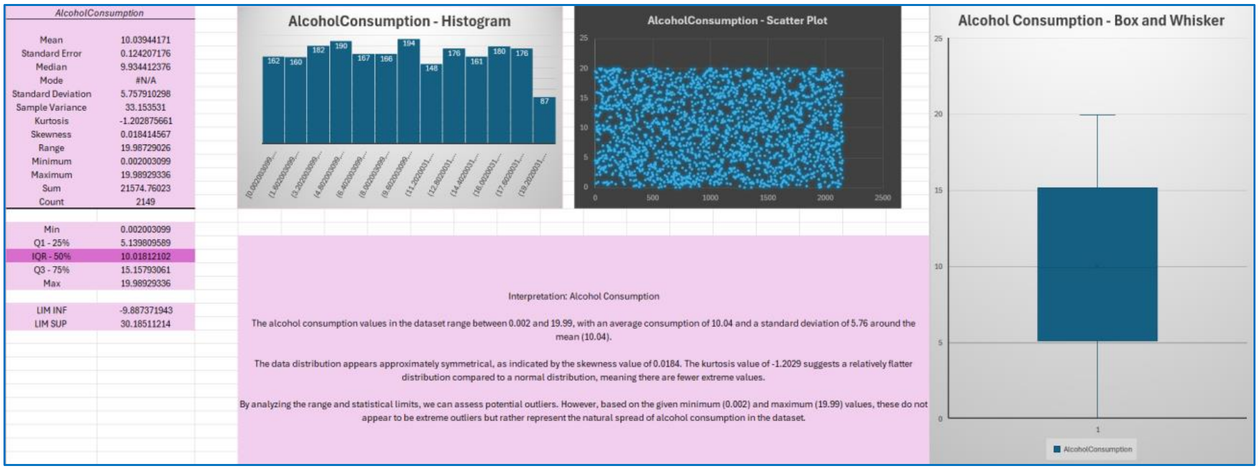
Interpretation: The BMI values in the dataset range between 15.01 and 39.99, with an average BMI of 27.66 and a standard deviation of 7.22 around the mean (27.66). The distribution of the data appears approximately symmetrical, as indicated by the skewness value of -0.0267. The kurtosis value of -1.1845 suggests a relatively flatter distribution compared to a normal distribution, meaning there are fewer extreme values. By analyzing the range and statistical limits, we can assess potential outliers. However, based on the given minimum (15.01) and maximum (39.99) values, these do not appear to be extreme outliers but rather represent the natural spread of BMI values in the dataset.



Datasource : Sheet BMI- excel file: Interpretations.xls

Alcohol Consumption: Weekly alcohol consumption in units, ranging from **0** to **20**.

Interpretation: The alcohol consumption values in the dataset range between 0.002 and 19.99, with an average consumption of 10.04 and a standard deviation of 5.76 around the mean (10.04). The data distribution appears approximately symmetrical, as indicated by the skewness value of 0.0184. The kurtosis value of -1.2029 suggests a relatively flatter distribution compared to a normal distribution, meaning there are fewer extreme values. By analyzing the range and statistical limits, we can assess potential outliers. However, based on the given minimum (0.002) and maximum (19.99) values, these do not appear to be extreme outliers but rather represent the natural spread of alcohol consumption in the dataset.

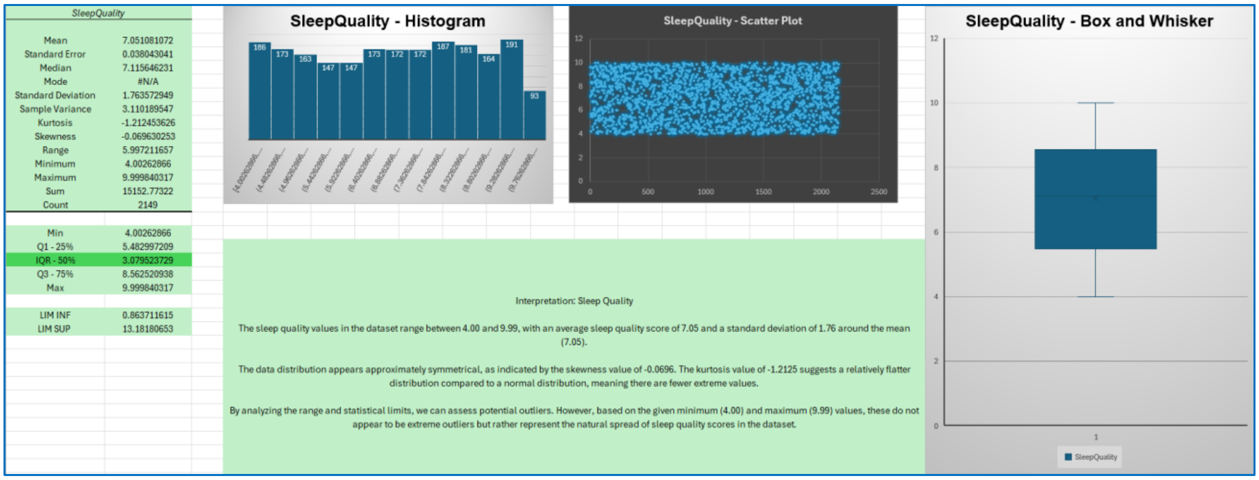


Datasource : Sheet Alcohol Consumption- excel file: Interpretations.xls

Sleep Quality: Sleep quality score, ranging from 4 to 10.

Interpretation: The sleep quality values in the dataset range between 4.00 and 9.99, with an average sleep quality score of 7.05 and a standard deviation of 1.76 around the mean (7.05). The data distribution appears approximately symmetrical, as indicated by the skewness value of -0.0696. The kurtosis value of -1.2125 suggests a relatively flatter distribution compared to a normal distribution, meaning there are fewer extreme values. By analyzing the range and statistical limits, we can assess potential outliers. However, based on the given minimum (4.00) and maximum (9.99) values, these do not appear to be extreme outliers but rather represent the natural spread of sleep quality scores in the dataset.

Predictive Analytics



Datasource : Sheet Sleep Quality- excel file: Interpretations.xls

Figure 1

Gender Distribution

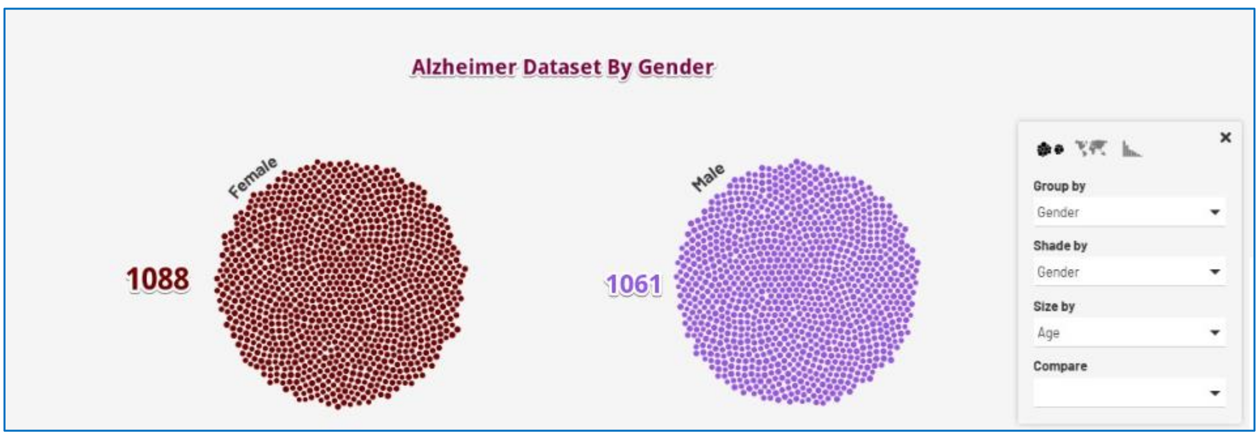


Figure 2

Age Distribution

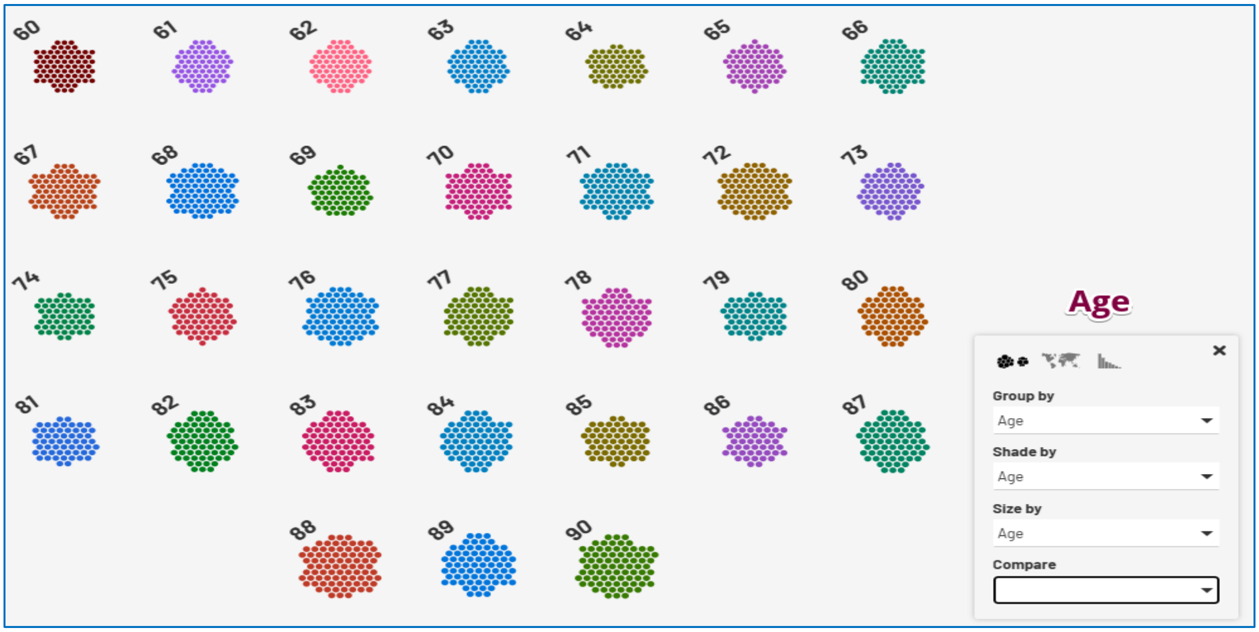


Figure 3

Smoker Distribution

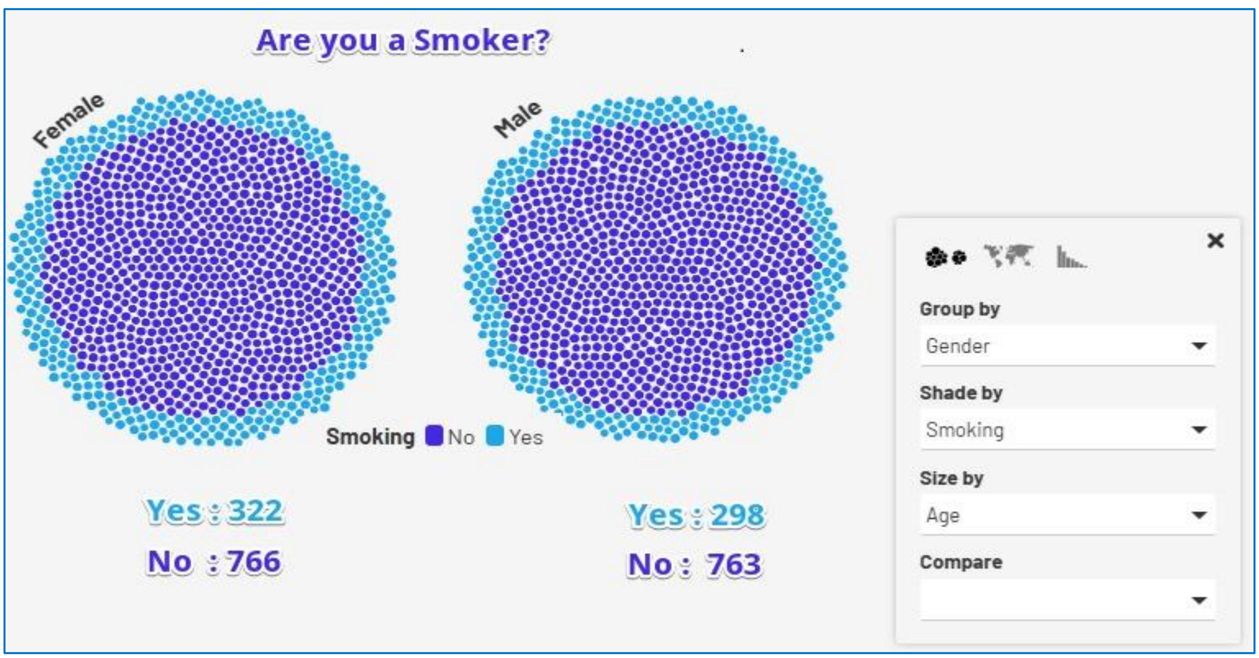


Figure 4

Ethnicity Distribution

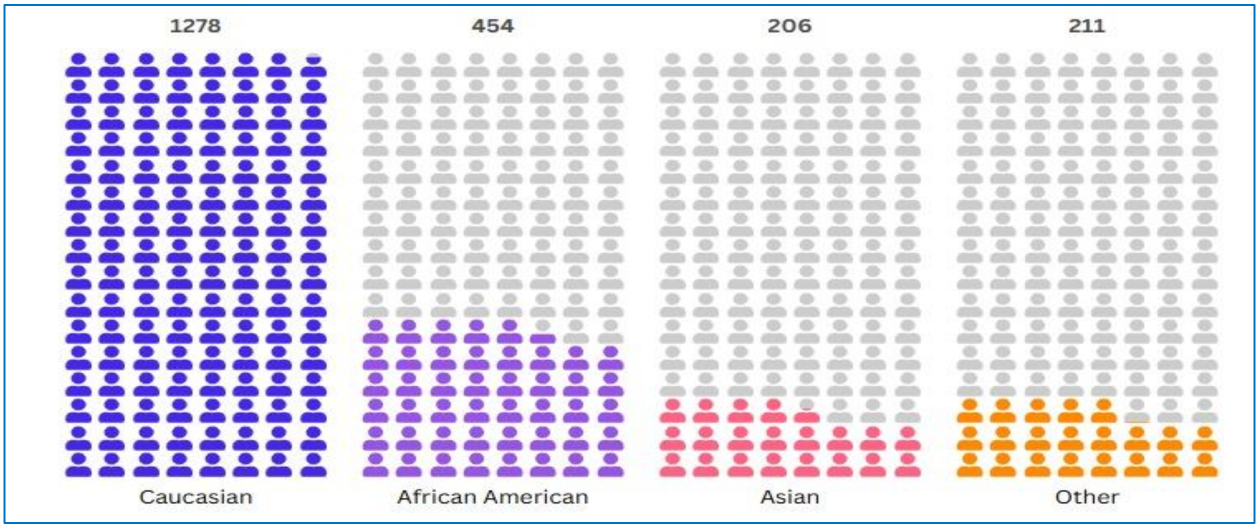
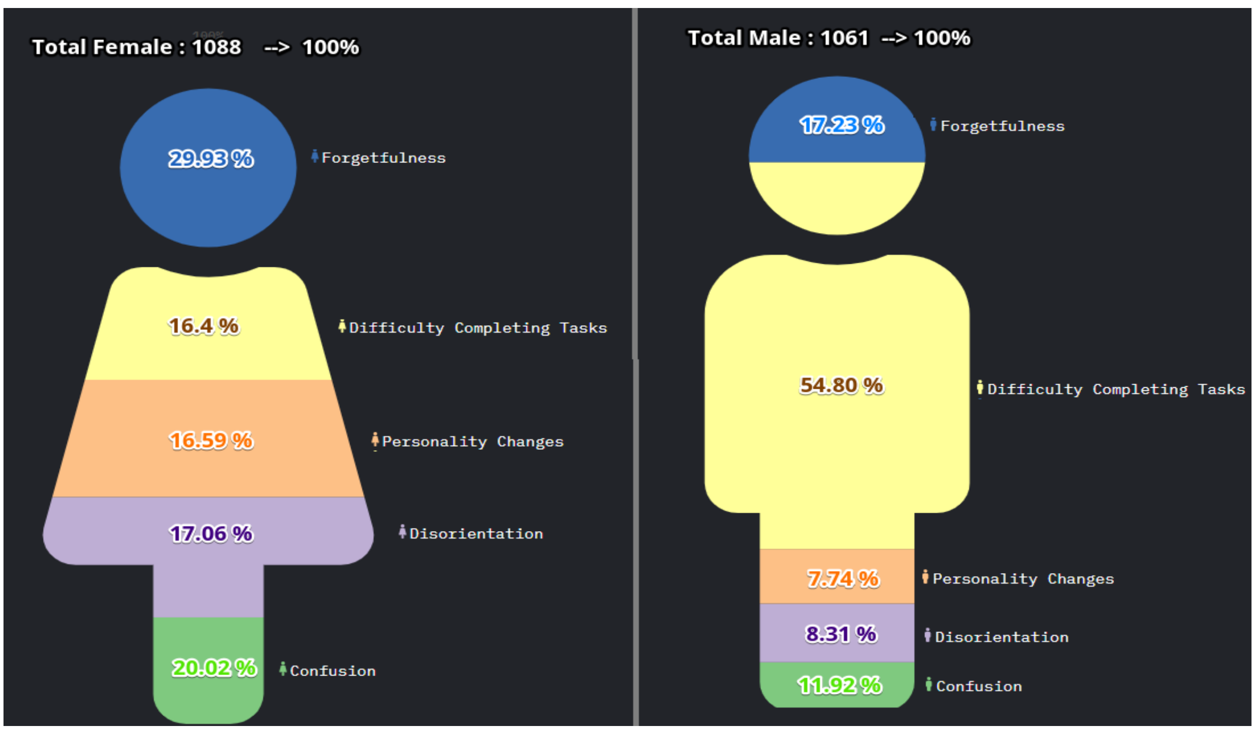


Figure 5

Segmentation by Male and Female



Note. Reference: https://public.flourish.studio/visualisation/22027739/

This graph represents the segmentation by Male and Female and how each of the symptoms is distributed throughout the entire population and at a simple glance we can see that although the number of men and women is similar, there is a symptom that has a higher percentage in men with 54.80%.

Cognitive and Functional Assessment

A decomposition tree has been created with Power BI to make the visualization easier to see the relationship of the cognitive variables, showing the number of people with positive diagnosis and to summarize variables such as ADL, Function Assessment and MMSE.

Figure 6

Cognitive relationships with negative Memory Complaint and negative Behavioral Problem.

Figure 7

Cognitive relationships with negative Memory Complaint and Positive Behavioral Problem.

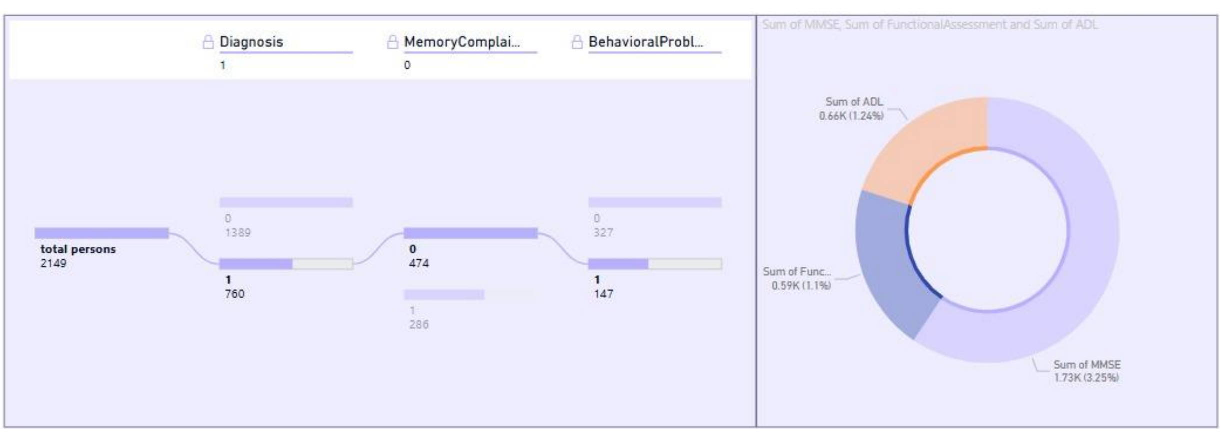


Figure 8

Cognitive relationships with positive Memory Complaint and negative Behavioral Problem.

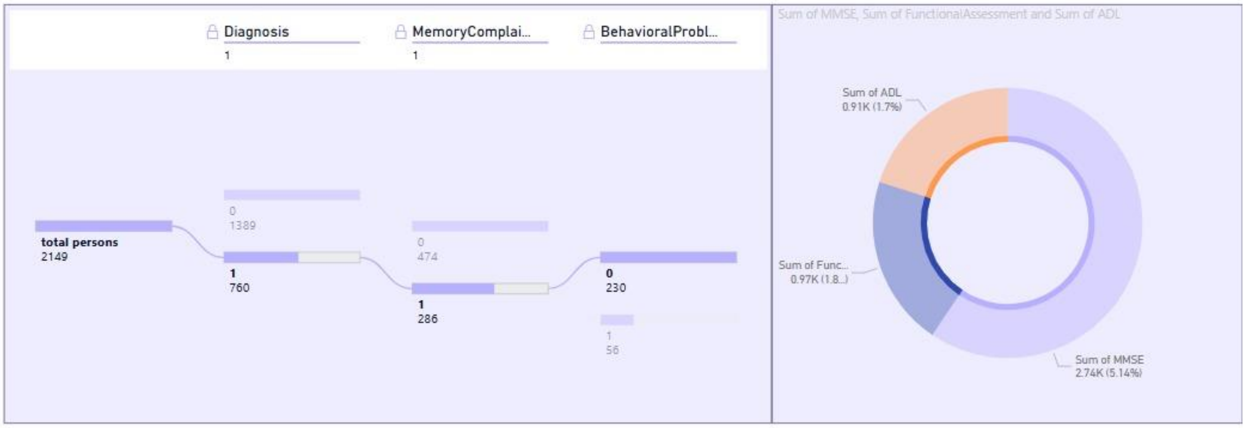
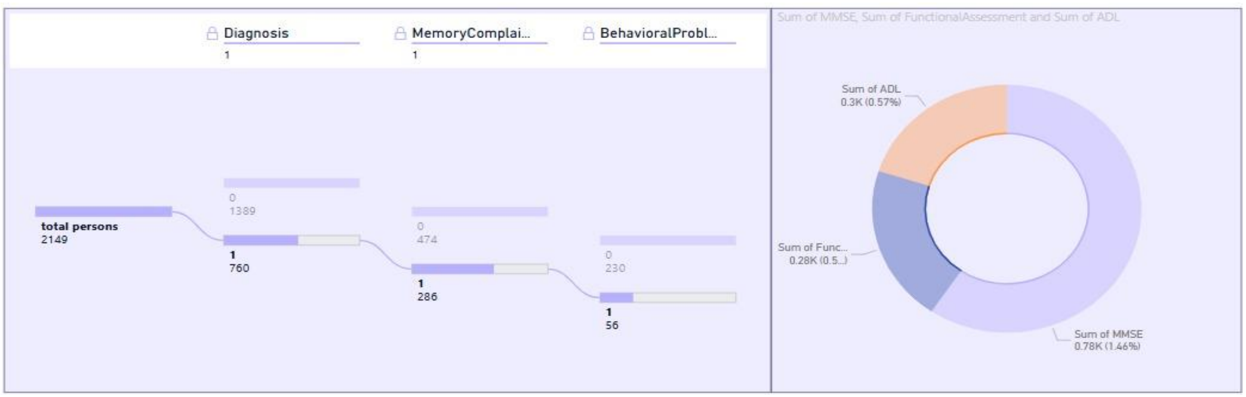


Figure 6

Cognitive relationships with positive Memory Complaint and positive Behavioral Problem.



Regression Model

As mentioned previously, our selected method for regression is the logistic model as we want to calculate the probability of a person having or not having Alzheimer's disease. To accomplish this, the first step was to do a correlation matrix to look at how the values behave when interacting with each other. The results from the matrix are shown next:

	Age	Gender	Etchinicity	EducationLevel	BMI	Smoking	AlcoholConsumption	PhysicalActivity	DietQuality	SleepQuality	FamilyHistoryAlzheimers	CardiovascularDisease	Diabetes	Depression	HeadInjury	Hypertension	SystolicBP	DiastolicBP	CholesterolTotal
Age	1.000000	0.028708	0.027745	-0.060560	-0.015677	0.020060	0.007323	-0.010249	-0.023386	0.049139	-0.018784	-0.015284	-0.005708	-0.008752	-0.040337	0.003954	-0.005324	-0.004462	0.000392
Gender	0.028708	1.000000	-0.003660	-0.007070	0.005928	0.016650	0.004330	-0.035795	0.010942	0.007310	0.018421	0.034577	-0.018300	0.018113	0.004013	-0.002641	0.011657	-0.026059	-0.009568
Etchinicity	0.027745	-0.003660	1.000000	0.026516	-0.003176	0.031485	0.008363	0.020498	-0.018184	0.029304	0.022526	0.002347	-0.019588	-0.005411	-0.009364	0.012848	-0.027086	0.010310	-0.011757
EducationLevel	-0.060560	-0.007070	0.026516	1.000000	-0.023514	-0.009904	-0.012646	-0.016703	0.017412	0.020248	0.032748	0.007528	0.003059	0.027573	-0.008951	-0.016950	-0.016782	-0.002553	-0.041598
BMI	-0.015677	0.005928	-0.003176	-0.023514	1.000000	0.020437	-0.008997	0.000742	0.019922	-0.005975	0.006446	-0.005467	-0.008980	-0.012507	0.016288	0.004928	-0.019275	-0.002524	0.001082
Smoking	0.020060	0.016650	0.031485	-0.009904	0.020437	1.000000	0.008363	0.010760	-0.002048	-0.001145	-0.045811	0.027955	-0.035810	-0.039363	-0.019178	-0.021124	-0.024116	-0.014112	-0.010907
AlcoholConsumption	0.007323	0.004330	0.008363	-0.012646	-0.008997	0.008363	1.000000	0.021696	0.020117	-0.003873	-0.003707	-0.023632	0.000073	0.008867	-0.008301	-0.006010	-0.030070	-0.008909	-0.033944
PhysicalActivity	-0.010249	-0.035795	0.020498	-0.016703	0.000742	0.010760	0.021696	1.000000	0.011085	-0.001823	-0.014106	0.003976	0.029880	-0.013533	0.025780	0.022081	-0.004811	-0.010555	0.014335
DietQuality	-0.023386	0.010942	-0.018184	0.017412	0.019922	-0.002048	0.020117	0.011085	1.000000	0.051295	-0.011261	-0.015220	0.009192	-0.002516	-0.008163	-0.043628	0.006031	0.009539	-0.016790
SleepQuality	0.049139	0.007310	0.029304	0.020248	-0.005975	-0.001145	-0.003873	-0.001823	0.051295	1.000000	0.014801	0.002112	0.023987	-0.022996	-0.004546	0.038668	-0.027887	0.010797	0.006879
FamilyHistoryAlzheimers	-0.018784	0.018421	0.022526	0.032748	0.006446	-0.045811	-0.003707	-0.014106	-0.011261	0.014801	1.000000	0.020783	-0.023106	0.000796	-0.019184	0.012920	0.006463	0.006060	0.007054
CardiovascularDisease	-0.015284	0.034577	0.002347	0.007528	-0.005467	0.027955	-0.023632	0.003976	-0.015220	0.002112	0.020783	1.000000	-0.010134	0.006043	0.001342	-0.011760	0.013575	-0.043662	-0.007961
Diabetes	-0.005708	-0.018300	-0.019588	0.003059	-0.008980	-0.035810	0.000073	0.029880	0.009192	0.023987	-0.023106	-0.010134	1.000000	0.003310	-0.000013	-0.000898	-0.011518	0.003423	0.048767
Depression	-0.008752	0.018113	-0.005411	0.027573	-0.012507	-0.039363	0.008867	-0.013533	-0.002516	-0.022996	0.000796	0.006043	0.003310	1.000000	-0.007662	0.032061	-0.018864	-0.023021	-0.010087
HeadInjury	-0.040337	0.004013	-0.009364	-0.008951	0.016288	-0.019178	-0.008301	0.025780	-0.008163	-0.004546	-0.019184	0.001342	-0.000013	-0.007662	1.000000	-0.016380	0.000081	0.010430	-0.021734
Hypertension	0.003954	-0.002641	0.012848	-0.016950	0.004928	-0.021124	-0.006010	0.022081	-0.043628	0.038668	0.012920	-0.011760	-0.000898	0.032061	-0.016380	1.000000	-0.000843	-0.043424	-0.024493
SystolicBP	-0.005324	0.011657	-0.027086	-0.016782	-0.019275	-0.024116	-0.030070	-0.004811	0.006031	-0.027887	0.006463	0.013575	-0.011518	-0.018864	0.000081	-0.000843	1.000000	0.002590	0.018537
DiastolicBP	-0.004462	-0.026059	0.010310	-0.002553	-0.002524	-0.014112	-0.008909	-0.010555	0.009539	0.010797	0.006060	-0.043662	0.003423	-0.023021	0.010430	-0.043424	0.002590	1.000000	0.015249
CholesterolTotal	0.000392	-0.009568	-0.011757	-0.041598	0.001082	-0.010907	-0.033944	0.014335	-0.016790	0.006879	0.007054	-0.007961	0.048767	-0.010087	-0.021734	-0.024493	0.018537	0.015249	1.000000
CholesterolLDL	0.003588	0.016271	0.007907	0.050504	0.023684	-0.007393	-0.017042	0.017789	-0.023698	0.006751	-0.011682	0.038607	0.014744	-0.004796	0.026806	0.013892	-0.007295	-0.015542	0.010336
CholesterolHDL	0.006803	-0.005846	-0.021828	-0.008125	0.038605	-0.025567	-0.001925	-0.001993	-0.008659	0.015453	-0.013581	-0.017681	0.038532	-0.008256	0.000685	-0.004205	0.002947	0.008180	0.010116
CholesterolTriglycerides	-0.003062	-0.012427	-0.007173	-0.025763	-0.018001	-0.021058	0.023362	0.027416	0.034734	0.024004	0.001046	-0.001264	-0.044906	0.043186	0.016845	-0.011504	-0.034824	-0.007928	-0.001959
MMSE	-0.004235	0.025330	-0.012504	0.026090	-0.003477	0.002139	-0.011424	-0.008342	0.021636	0.012506	0.015575	0.025351	-0.008905	0.014020	0.006063	0.012079	-0.003548	-0.028586	-0.013638
FunctionalAssessment	0.005508	0.033324	-0.004704	0.020269	-0.030501	-0.031879	-0.016483	-0.002419	-0.009700	0.029589	-0.004650	-0.041270	0.043548	0.019733	0.034655	-0.028332	0.013161	0.031485	-0.005345
MemoryComplaints	0.012343	0.003880	0.003687	-0.000165	0.029293	0.020339	-0.029418	0.008709	0.012537	-0.016807	-0.028348	0.031061	-0.017280	-0.019039	-0.021329	-0.021130	-0.009517	-0.006126	0.013821
BehavioralProblems	0.038973	0.006099	-0.019363	0.010473	0.035276	-0.014763	0.013657	-0.014253	-0.014033	-0.019899	-0.023558	-0.016803	-0.027927	0.001316	0.047649	0.031700	-0.012787	0.016682	-0.006505
ADL	-0.038658	0.003865	0.010475	0.031934	-0.009788	-0.029233	-0.008250	-0.013324	-0.007888	0.014535	0.014041	-0.012918	0.022235	0.006195	-0.016669	-0.018602	0.015447	-0.004836	0.000775
Confusion	0.009002	-0.030583	0.022431	-0.008168	-0.015859	0.004498	-0.032798	-0.009397	0.002628	0.013508	0.017976	0.017658	-0.004793	0.001593	-0.031152	-0.008634	0.008404	-0.016567	0.037390
Disorientation	0.027523	0.017506	-0.018132	-0.017568	-0.031840	-0.028403	0.015521	-0.020407	-0.026271	0.019467	0.035963	0.025237	-0.004494	0.005764	0.033060	0.033567	0.037500	-0.015098	0.025313
PersonalityChanges	-0.009470	0.025919	-0.030035	-0.019950	-0.015901	-0.009977	0.019117	-0.013629	0.029087	-0.016892	0.006839	-0.032343	0.004184	0.026044	-0.031417	-0.011857	-0.012170	0.001501	-0.017469
DifficultyCompletingTasks	0.013900	-0.001636	0.019363	0.013033	-0.039400	0.001741	-0.002712	0.031169	0.045737	0.011237	0.008787	0.024684	-0.001466	-0.017145	0.010646	-0.002780	-0.003598	-0.026220	-0.017936
Forgetfulness	-0.019279	-0.028537	-0.034619	-0.008685	0.071131	0.018011	-0.024121	0.009837	0.006175	0.002930	0.022339	0.010170	-0.007644	0.017822	-0.000020	0.029931	-0.011357	0.005857	-0.025326
Diagnosis	-0.005488	-0.020975	-0.014782	-0.043966	0.026343	-0.004865	-0.007618	0.005945	0.008506	-0.056548	-0.032900	0.031490	-0.031508	-0.005893	-0.021411	0.035080	-0.015615	0.005293	0.006394

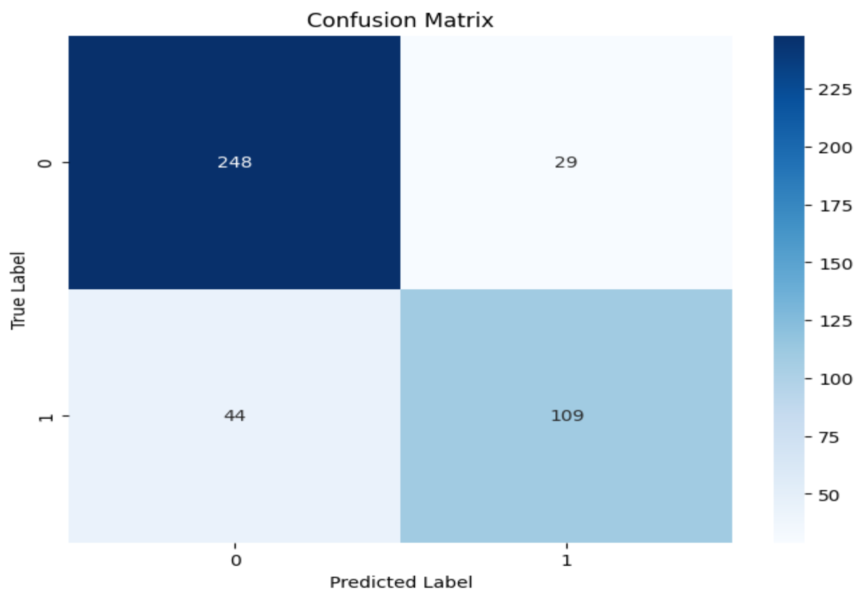
CholesterolTotal	CholesterolLDL	CholesterolHDL	CholesterolTriglycerides	MMSE	FunctionalAssessment	MemoryComplaints	BehavioralProblems	ADL	Confusion	Disorientation	PersonalityChanges	DifficultyCompletingTasks	Forgetfulness	Diagnosis
0.000392	0.003588	0.006803	-0.003062	-0.004235	0.005508	0.012343	0.038973	-0.038658	0.009002	0.027523	-0.009470	0.013900	-0.019279	-0.005488
-0.009568	0.016271	-0.005846	-0.012427	0.025330	0.033324	0.003880	0.006099	0.003865	-0.030583	0.017506	0.025919	-0.001636	-0.028537	-0.020975
-0.011757	0.007907	-0.021828	-0.007173	-0.012504	-0.004704	0.003687	-0.019363	0.010475	0.022431	-0.018132	-0.030035	0.019363	-0.034619	-0.014782
-0.041598	0.050504	-0.008125	-0.025763	0.026090	0.020269	-0.000165	0.010473	0.031934	-0.008168	-0.017568	-0.019950	0.013033	-0.008685	-0.043966
0.001082	0.023684	0.038605	-0.018001	-0.003477	-0.030501	0.029293	0.035276	-0.009788	-0.015859	-0.031840	-0.015901	-0.039400	0.071131	0.026343
-0.010907	-0.007393	-0.025567	-0.021058	0.002139	-0.031879	0.020339	-0.014763	-0.029233	0.004498	-0.028403	-0.009977	0.001741	0.018011	-0.004865
-0.033944	-0.017042	-0.001925	0.023362	-0.011424	-0.016483	-0.029418	0.013657	-0.008250	-0.032798	0.015521	0.019117	-0.002712	-0.024121	-0.007618
0.014335	0.017789	-0.001993	0.027416	-0.008342	-0.002419	0.008709	-0.014253	-0.013324	-0.009397	-0.020407	-0.013629	0.031169	0.009837	0.005945
-0.016790	-0.023698	-0.008659	0.034734	0.021636	-0.009700	0.012537	-0.014033	-0.007888	0.002628	-0.026271	0.029087	0.045737	0.006175	0.008506
0.006879	0.006751	0.015453	0.024004	0.012506	0.029589	-0.016807	-0.019899	0.014535	0.013508	0.019467	-0.016892	0.011237	0.002930	-0.056548
0.007054	-0.011682	-0.013581	0.001046	0.015575	-0.004650	-0.028348	-0.023558	0.014041	0.017976	0.035963	0.006839	0.008787	0.022339	-0.032900
-0.007961	0.038607	-0.017681	-0.001264	0.025351	-0.041270	0.031061	-0.016803	-0.012918	0.017658	0.025237	-0.032343	0.024684	0.010170	0.031490
0.048767	0.014744	0.038532	-0.044906	-0.008905	0.043548	-0.017280	-0.027927	0.022235	-0.004793	-0.004494	0.004184	-0.001466	-0.007644	-0.031508
-0.010087	-0.004796	-0.008256	0.043186	0.014020	0.019733	-0.019039	0.001316	0.006195	0.001593	0.005764	0.026044	-0.017145	0.017822	-0.005893
-0.021734	0.026806	0.000685	0.016845	0.006063	0.034655	-0.021329	0.047649	-0.016669	-0.031152	0.033060	-0.031417	0.010646	-0.000020	-0.021411
-0.024493	0.013892	-0.004205	-0.011504	0.012079	-0.028332	-0.021130	0.031700	-0.018602	-0.008634	0.033567	-0.011857	-0.002780	0.029931	0.035080
0.018537	-0.007295	0.002947	-0.034824	-0.003548	0.013161	-0.009517	-0.012787	0.015447	0.008404	0.037500	-0.012170	-0.003598	-0.011357	-0.015615
0.015249	-0.015542	0.008180	-0.007928	-0.028586	0.031485	-0.006126	0.016682	-0.004836	-0.016567	-0.015098	0.001501	-0.026220	0.005857	0.005293
1.000000	0.010336	0.010116	-0.001959	-0.013638	-0.005345	0.013821	-0.006505	0.000775	0.037390	0.025313	-0.017469	-0.017936	-0.025326	0.006394
0.010336	1.000000	-0.037148	-0.005582	0.025383	-0.016215	-0.005737	-0.002926	-0.019380	0.020629	0.027335	0.016120	-0.011552	-0.000588	-0.031976
0.010116	-0.037148	1.000000	0.015235	-0.005218	-0.003450	0.030898	0.027170	0.005252	-0.005254	0.021054	-0.018553	-0.056947	-0.016662	0.042584
-0.001959	-0.005582	0.015235	1.000000	-0.007615	-0.009207	-0.014484	0.010310	0.023546	-0.013631	-0.013222	-0.017733	-0.041899	0.003526	0.022672
-0.013638	0.025383	-0.005218	-0.007615	1.000000	0.024932	0.007652	0.025408	0.003359	0.003763	0.036715	0.034492	0.003294	0.011848	-0.237126
-0.005345	-0.016215	-0.003450	-0.009207	0.024932	1.000000	0.002320	-0.021941	0.053904	-0.023118	-0.007193	0.024260	0.019805	0.019387	-0.364898
0.013821	-0.005737	0.030898	-0.014484	0.007652	0.002320	1.000000	-0.009765	-0.037511	-0.007749	0.013442	-0.030096	0.044151	-0.006961	0.306742
-0.006505	-0.002926	0.027170	0.010310	0.025408	-0.021941	-0.009765	1.000000	0.043376	-0.022676	-0.022154	-0.002892	-0.019174	0.023374	0.224350
0.000775	-0.019380	0.005252	0.023546	0.003359	0.053904	-0.037511	0.043376	1.000000	0.012373	0.002256	-0.016128	-0.025646	-0.002595	-0.332346
0.037390	0.020629	-0.005254	-0.013631	0.003763	-0.023118	-0.007749	-0.022676	0.012373	1.000000	0.007034	-0.001573	-0.022003	0.010100	-0.019186
0.025313	0.027335	0.021054	-0.013222	0.036715	-0.007193	0.013442	-0.022154	0.002256	0.007034	1.000000	-0.022311	-0.020767	-0.032013	-0.024648
-0.017469	0.016120	-0.018553	-0.017733	0.034492	0.024260	-0.030096	-0.002892	-0.016128	-0.001573	-0.022311	1.000000	0.037686	-0.013311	-0.020627
-0.017936	-0.011552	-0.056947	-0.041899	0.003294	0.019805	0.044151	-0.019174	-0.025646	-0.022003	-0.020767	0.037686	1.000000	-0.013386	0.009069
-0.025326	-0.000588	-0.016662	0.003526	0.011848	0.019387	-0.006961	0.023374	-0.002595	0.010100	-0.032013	-0.013311	-0.013386	1.000000	-0.000354
0.006394	-0.031976	0.042584	0.022672	-0.237126	-0.364898	0.306742	0.224350	-0.332346	-0.019186	-0.024648	-0.020627	0.009069	-0.000354	1.000000

As it can be seen in the matrix most of the variables don’t have high correlation values with Diagnosis (dependent variable) except for 5 variables which curiously belong to the cognitive and functional assessments category. However, just to be sure it was decided to include all the variables to see the model performance, but just to make sure we added the function to check for high correlation between the independent variables not to violate one of the principles of logistic regression. Then, the model was executed, and the results are the following:

```
Accuracy: 0.83
Classification Report:
              precision    recall  f1-score   support

           0       0.85      0.90      0.87       277
           1       0.79      0.71      0.75       153

    accuracy                           0.83       430
   macro avg       0.82      0.80      0.81       430
weighted avg       0.83      0.83      0.83       430
```




```
Optimization terminated successfully.
         Current function value: 0.364545
         Iterations 7
                           Logit Regression Results
==============================================================================
Dep. Variable:              Diagnosis   No. Observations:                 1719
Model:                          Logit   Df Residuals:                     1686
Method:                           MLE   Df Model:                           32
Date:                Mon, 10 Mar 2025   Pseudo R-squ.:                  0.4386
Time:                        20:22:58   Log-Likelihood:                -626.65
converged:                       True   LL-Null:                       -1116.2
Covariance Type:            nonrobust   LLR p-value:                4.180e-185
```

Dep. Variable:	Diagnosis	No. Observations:	1719
Model:	Logit	Df Residuals:	1686
Method:	MLE	Df Model:	32
Date:	Mon, 10 Mar 2025	Pseudo R-squ.:	0.4386
Time:	20:22:58	Log-Likelihood:	-626.65
converged:	True	LL-Null:	-1116.2
Covariance Type:	nonrobust	LLR p-value:	4.180e-185

	coef	std err	z	P>|z|	[0.025	0.975]
const	4.9907	1.088	4.586	0.000	2.858	7.124
Age	-0.0132	0.008	-1.682	0.092	-0.029	0.002
Gender	0.1269	0.144	0.884	0.377	-0.155	0.408
Etchinicity	-0.0216	0.073	-0.296	0.767	-0.165	0.121
EducationLevel	-0.1465	0.080	-1.838	0.066	-0.303	0.010
BMI	-0.0049	0.010	-0.489	0.625	-0.024	0.015
Smoking	-0.2506	0.161	-1.557	0.119	-0.566	0.065
AlcoholConsumption	-0.0054	0.012	-0.438	0.661	-0.029	0.019
PhysicalActivity	-0.0328	0.025	-1.306	0.192	-0.082	0.016
DietQuality	0.0247	0.025	0.984	0.325	-0.025	0.074
SleepQuality	-0.0537	0.041	-1.306	0.192	-0.134	0.027
FamilyHistoryAlzheimers	-0.1149	0.167	-0.688	0.491	-0.442	0.212
CardiovascularDisease	0.0697	0.201	0.347	0.728	-0.323	0.463
Diabetes	0.2081	0.203	1.025	0.306	-0.190	0.606
Depression	0.2000	0.172	1.164	0.244	-0.137	0.537
HeadInjury	-0.3369	0.244	-1.383	0.167	-0.814	0.141
Hypertension	0.1577	0.197	0.799	0.424	-0.229	0.545
SystolicBP	-0.0003	0.003	-0.118	0.906	-0.006	0.005
DiastolicBP	0.0054	0.004	1.358	0.174	-0.002	0.013
CholesterolTotal	0.0001	0.002	0.072	0.942	-0.003	0.003
CholesterolLDL	-0.0015	0.002	-0.915	0.360	-0.005	0.002
CholesterolHDL	0.0053	0.003	1.722	0.085	-0.001	0.011
CholesterolTriglycerides	0.0009	0.001	1.299	0.194	-0.000	0.002
MMSE	-0.1023	0.009	-11.200	0.000	-0.120	-0.084
FunctionalAssessment	-0.4730	0.030	-15.586	0.000	-0.533	-0.414
MemoryComplaints	2.6859	0.190	14.125	0.000	2.313	3.059
BehavioralProblems	2.4209	0.207	11.690	0.000	2.015	2.827
ADL	-0.4413	0.030	-14.799	0.000	-0.500	-0.383
Confusion	-0.1758	0.180	-0.975	0.330	-0.529	0.178
Disorientation	0.0169	0.197	0.086	0.932	-0.369	0.403
PersonalityChanges	-0.0935	0.206	-0.455	0.649	-0.497	0.309
DifficultyCompletingTasks	-0.0020	0.196	-0.010	0.992	-0.386	0.382
Forgetfulness	0.0956	0.157	0.608	0.543	-0.213	0.404

The results are interesting and will be explained here: First the R-squared has a value of 43.86% which is not the most optimal value as all the variables in the database only explain this percentage of Diagnosis variation. Next, the LLR p-value indicates that the model is statistically significant as its value is almost zero. The next part to analyze are the p-values of each variable which as it can be seen is that most of the variables are not statistically significant except for the 5 that belong to the Cognitive and Functional Assessments category which are statistically significant for the model. The closest variables to the 0.05 threshold are Age, BMI, and CholesterolHDL which means that these variables explain the outcome of Alzheimer disease, but in a very little amount.

These 5 variables are important as their coefficients explain that lower scores on the Mini-Mental State Exam (MMSE) may belong to people with Alzheimer, then Functional Assessment has a negative value meaning that decrease functions are linked to the condition. Memory complaints is positive and shows that having these changes are related with the disease. Having behavioral problems is also a characteristic that Alzheimer's patients have which explains the positive relationship. Finally, ADL or activity of daily living was a measure from 1 to 10 where low values are linked with Alzheimer's disease.

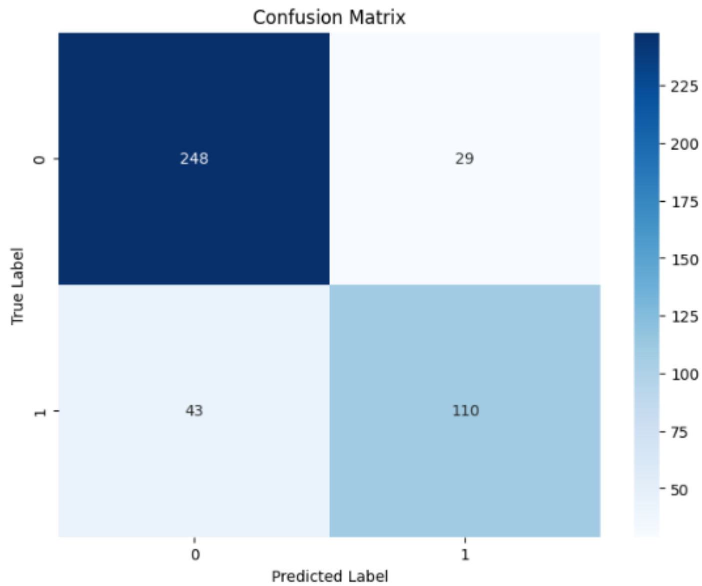
The next part to analyze are the accuracy, precision and recall. The general accuracy for the model is 83% which is a good value considering the many variables that are not favorable for the model. The precision and recall values are good for identifying non-Alzheimer's cases with values of 0.85 and 0.90 respectively. On the other hand, the precision and recall values are not that optimal for positively identifying Alzheimer patients with values of 0.79 and 0.71 respectively. Finally, the confusion matrix shows that there are more true positives and negatives than the false ones, however the number of missed cases is still high which means that the model may need optimization.

It was also seen that a second model could be implemented where only the 5 important variables would be considered to do the regression with the following results:

```
Accuracy: 0.83
Classification Report:
              precision    recall  f1-score   support

           0       0.85      0.90      0.87       277
           1       0.79      0.72      0.75       153

    accuracy                           0.83       430
   macro avg       0.82      0.81      0.81       430
weighted avg       0.83      0.83      0.83       430
```




```
Optimization terminated successfully.
         Current function value: 0.372994
         Iterations 7
                           Logit Regression Results
==============================================================================
Dep. Variable:              Diagnosis   No. Observations:                 1719
Model:                          Logit   Df Residuals:                     1713
Method:                           MLE   Df Model:                            5
Date:                Mon, 10 Mar 2025   Pseudo R-squ.:                  0.4256
Time:                        20:33:50   Log-Likelihood:                -641.18
converged:                       True   LL-Null:                       -1116.2
Covariance Type:            nonrobust   LLR p-value:                3.776e-203
==========================================================================================
                             coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                      3.9072      0.260     15.016      0.000       3.397       4.417
MMSE                      -0.1014      0.009    -11.374      0.000      -0.119      -0.084
FunctionalAssessment      -0.4613      0.029    -15.718      0.000      -0.519      -0.404
MemoryComplaints           2.6500      0.187     14.194      0.000       2.284       3.016
BehavioralProblems         2.3733      0.202     11.721      0.000       1.976       2.770
ADL                       -0.4261      0.029    -14.770      0.000      -0.483      -0.370
==========================================================================================
```

The second model shows very similar characteristics with the first model. First, it has a lower R-squared at 42.56% which may be due to the other variables not included in the model, but the decrease is not as big, so the used variables still predict a good amount of the Diagnosis variable. Next, the LLR p-value is almost zero which is a good indicator for the model. The p-values for the variables show that all of them are statistically significant. Finally, the accuracy of the model didn't increase nor decrease, similarly with the precision and recall of the model. The confusion matrix only added 1 true positive and took 1 false negative. Therefore, this model is like the first one and only is better in few areas.

Conclusions

From these models we can clearly say that having problems in the functional and cognitive areas are a clear indicator that a person may have Alzheimer. Other variables such as age, or Body Mass Index (BMI) may be a sign but are not high enough to be considered when diagnosing a person with this condition. Moreover, age is always considered to be an indicator as seniors are the people that have this condition.

It is also recommended that further study needs to be conducted to clearly identify what characteristics, patterns, behavior and so on may be an indicator of a person having Alzheimer as using this data would surely improve the accuracy of the model used here.

REFERENCES

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